



46755 - Renewables in Electricity Markets

ASSIGNMENT B - GROUP 13

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Table of contributions

Table of Contributions	s243099	s243102	s223862	s223848
Contribution to model development	25%	25%	25%	25%
Contribution to programming	25%	25%	25%	25%
Contribution to the analysis of numerical results	25%	25%	25%	25%
Contribution to writing the report	25%	25%	25%	25%

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Nomenclature

Sets:

$t \in T = \{1, 2, \dots, 24\}$:= set of time-frames

$\omega \in \Omega = \{1, 2, \dots, 1600\}$:= set of scenarios

$m \in \mathcal{M} = \{1, 2, \dots, 60\}$:= set of minutes in the bidding hour

$w \in W = \{1, 2, \dots, |W|\}$:= set of scenarios (load profiles)

Parameters:

$\pi_\omega = \frac{1}{1600}$:= probability of each scenario

$P^{nom} = 500$:= Nominal production capacity of the wind farm [MW]

$\lambda_{t\omega}^{DA}$:= day-ahead price in hour t , scenario ω [€/MWh]

$P_{t\omega}^{real}$:= realized wind power production in hour t , scenario ω [MW]

$SI_{t\omega} \in \{0, 1\}$:= system imbalance indicator (1 = supply deficit, 0 = supply surplus)

$\lambda_{t\omega}^{BP}$:= balancing price in hour t , scenario ω , and it's defined as follows:

$$\lambda_{t\omega}^{BP} = \begin{cases} 1.25 \times \lambda_{t\omega}^{DA} & \text{if } SI_{t\omega} = 1 \\ 0.85 \times \lambda_{t\omega}^{DA} & \text{if } SI_{t\omega} = 0 \end{cases}$$

Π_ω := total profit realized in scenario ω [EUR]

$\zeta \in R$:= Value-at-Risk threshold, determined endogenously by the optimization [EUR]

$\eta_\omega \geq 0$:= profit shortfall below ζ in scenario ω , i.e. $\eta_\omega = \max\{0, \zeta - \Pi_\omega\}$ [EUR]

$\beta \geq 0$:= risk-aversion weight controlling the trade-off between expected profit and CVaR (parameter set by the decision maker)

$\alpha = 0.90$:= CVaR confidence level; the CVaR corresponds to the expected profit in the worst $(1 - \alpha) \times 100\% = 10\%$ of scenarios

$c^\uparrow$:= FCR-D UP reserve bid (kW)

$F_{m,w}^\uparrow$:= available upward flexibility at minute m under scenario w [kW]

$y_{m,w}$:= binary variable equal to 1 if the reserve constraint is violated at minute m under scenario w , 0 otherwise

$\zeta_{m,w}$:= auxiliary variable representing excess shortfall at minute m under scenario w [kW]

β := CVaR auxiliary threshold variable (Value-at-Risk proxy)

$\varepsilon = 0.1$:= allowed violation probability in the case of P90 requirement

M := Big-M used for ALSO-X formulation chosen sufficiently large, e.g. $M = 600$

$L_{m,w}$:= load at minute m under scenario w [kW]

$q = \varepsilon|M||W|$:= admissible violation budget, used in both ALSO-X and CVaR formulations

Decision variables:

p_t^{DA} := optimal offering quantity in the day-ahead market in hour t [MW]

Auxiliary variables:

$\Delta_{t\omega}^\uparrow$:= upward deviation (excess production) in hour t , scenario ω [MW]

$\Delta_{t\omega}^\downarrow$:= downward deviation (deficit production) in hour t , scenario ω [MW]

Use of AI

In accordance with DTU guidelines, the group used generative AI through ChatGPT, specifically GPT-5.4, to assist with coding, troubleshooting, text summarization, and grammar correction. AI served as support for the writing and development process, while the contents, interpretation of results and final conclusions were derived independently by the group.

Introduction

This report addresses different scenarios in which a wind farm may operate. The specific conditions under which these scenarios have been constructed are presented throughout each section.

As requested by the problem statement, the baseline data sources are as follows:

- **Wind Power Production Forecast:** Data were extracted from Renewables.ninja [2] — specifically, the *Renewables.ninja Wind (current onshore fleet, country zones, hourly, 1980–2024)* dataset for Denmark (`ninja_wind_countryzones_hourly_DK_current-onshore_merra2.csv`), Version 1.4, licensed under CC BY-NC 4.0, with methodology described in [3]. The dataset provides hourly bias-corrected capacity factors in the range $[0, 1]$, representing a simulated installed capacity of 4,773 MW across multiple Danish NUTS2 zones. Since the wind farm is assumed to be located in the DK2 bidding zone, only the DK02 column was retained, and additional data processing was required to extract and prepare this subset for use in the scenario generation. The wind power generation data cover the period from 2024-12-12 00:00 UTC to 2024-12-31 23:00 UTC.
- **Day-Ahead Price Forecast:** Historical hourly day-ahead prices were extracted from the Nord Pool Group [1] for the DK2 bidding zone, covering the period from 1 March to 20 March, with prices denominated in EUR.

It should be noted that the two data sources cover different time periods: the wind power production data span the period from 2024-12-12 to 2024-12-31, while the day-ahead price data cover 1 March to 20 March. This mismatch prevents a direct temporal alignment between the two sources, which may limit the representativeness of their joint scenarios. However, this choice was deliberate: rather than forcing a comparison grounded in a shared historical baseline, the focus of this report is on assessing the behaviour and performance of the proposed offering strategies. The results are therefore evaluated in terms of structural properties and decision quality, rather than on the fidelity of the underlying data to a specific real-world period.

Step 1 - Participation in Day-ahead and Balancing Markets

Task 1.1

Under a one-price scheme the mathematical model has been written as follows, please consider that for all the mathematical from this section on, the models will present solely the nomenclature of variables and parameters, their actual values are reported in the Nomenclature section at the beginning of the report.

Mathematical Model (1)

- **Objective function**

$$\max \sum_{t \in T} \sum_{\omega \in \Omega} \pi_{\omega} \left[\lambda_{t\omega}^{\text{DA}} p_t^{\text{DA}} + \text{SI}_{t\omega} \lambda_{t\omega}^{\text{BP}} (\Delta_{t\omega}^{\uparrow} - \Delta_{t\omega}^{\downarrow}) + \lambda_{t\omega}^{\text{BP}} (1 - \text{SI}_{t\omega}) (\Delta_{t\omega}^{\uparrow} - \Delta_{t\omega}^{\downarrow}) \right] \quad (1)$$

- **Constraints**

$$0 \leq p_t^{\text{DA}} \leq P^{\text{nom}} \quad \forall t \in T \quad (2)$$

$$\Delta_{t\omega}^{\uparrow} - \Delta_{t\omega}^{\downarrow} = p_{t\omega}^{\text{real}} - p_t^{\text{DA}} \quad \forall t \in T, \forall \omega \in \Omega \quad (3)$$

$$0 \leq \Delta_{t\omega}^{\uparrow} \leq P^{\text{nom}} \quad \forall t \in T, \forall \omega \in \Omega \quad (4)$$

$$0 \leq \Delta_{t\omega}^{\downarrow} \leq P^{\text{nom}} \quad \forall t \in T, \forall \omega \in \Omega \quad (5)$$

(2)

Assumptions and considerations

The two balancing terms in the objective function, $SI_{t\omega}\lambda_{t\omega}^{BP}(\Delta_{t\omega}^{\uparrow} - \Delta_{t\omega}^{\downarrow})$ and $(1 - SI_{t\omega})\lambda_{t\omega}^{BP}(\Delta_{t\omega}^{\uparrow} - \Delta_{t\omega}^{\downarrow})$, are kept explicit to reflect the physical structure of the one-price scheme: the settlement price $\lambda_{t\omega}^{BP}$ is determined solely by the system state $SI_{t\omega}$, and is applied symmetrically to both upward and downward deviations of the wind farm, regardless of whether its imbalance is beneficial or harmful to the system. This stands in direct contrast to the two-price scheme in Task 1.2, where the settlement price differs depending on the interaction between the system state and the direction of the wind farm's own imbalance.

Hence, for simplicity the objective function could also be written in a compact form as:

$$\max \sum_{t \in T} \sum_{\omega \in \Omega} \pi_{\omega} \left[\lambda_{t\omega}^{DA} p_t^{DA} + \lambda_{t\omega}^{BP} (\Delta_{t\omega}^{\uparrow} - \Delta_{t\omega}^{\downarrow}) \right]$$

Results

The optimal hourly day-ahead offers is largely influenced by the outcome of the imbalance scenario. During a one-price scheme for the balancing, the balancing strategy tend to go for extreme bids either bidding 0 MW or 500 MW (max capacity). Since all imbalances are settled at the balancing price, regardless of whether it is beneficial or harmful to the system. As a result of this, the producer can use the price difference between the DA market and the balancing market without being penalized. It results in the offering strategy that the producer either offers 0 or 500 MW in the balancing market, strongly dependent on whether the system is expected to be in deficit or surplus. The results are presented in Table 1.

Hour	1	2	3	4	5	6	7	8	9	10	11	12
Offer [MW]	0	500	0	0	500	0	0	0	0	0	500	0

Hour	13	14	15	16	17	18	19	20	21	22	23	24
Offer [MW]	0	500	0	0	0	0	0	0	0	0	500	0

Table 1: Optimal offers for each hour across all simulated scenarios

Using this bidding strategy, wind generation is assumed to have a marginal production cost of zero. As a result, all revenues can be interpreted directly as profits. The distribution of profits across all simulated scenarios under the one-price scheme is illustrated in Figure 1. The figure shows the variability in outcomes driven by the exposure to balancing market prices and the producer's optimal bidding behavior.

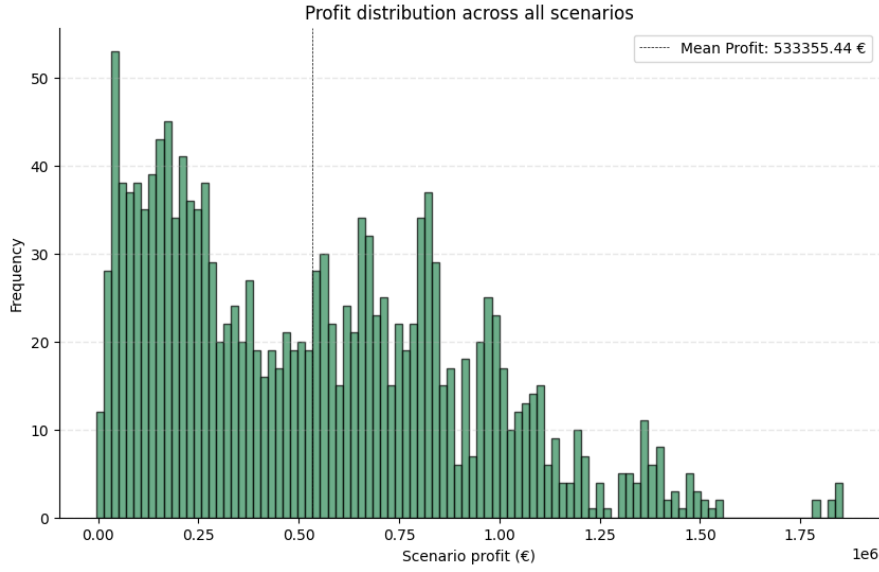


Figure 1: Distribution of profits across all scenarios w. one price scheme

The profit distribution shows a wide and generally positive generating profit histogram. The range of outcomes lies between values close to zero up to approximately 1.8€ million. A significant concentration of scenarios is clustered in the lower range of profits between zero and 0.25€ million. The distribution shows that larger profits are made, but are more rare in comparison to the low profit outcomes. The mean profit indicated through the dashed line, which lies on the better side of 0.5€ million.

Task 1.2

Mathematical Model (2)

The model from the previous task has been extended to a two-price scheme. Here, the settlement price depends on the interaction between the system state $SI_{t\omega} = 1$ and the direction of the wind farm's imbalance, distinguishing between beneficial and harmful deviations.

- **Objective function**

$$\max_{p_t^{DA}, \Delta_{t\omega}^{\uparrow}, \Delta_{t\omega}^{\downarrow}} \sum_{t \in T} \sum_{\omega \in \Omega} \pi_{\omega} \left[\lambda_{t\omega}^{DA} p_t^{DA} + SI_{t\omega} (\lambda_{t\omega}^{DA} \Delta_{t\omega}^{\uparrow} - \lambda_{t\omega}^{BP} \Delta_{t\omega}^{\downarrow}) + (1 - SI_{t\omega}) (\lambda_{t\omega}^{BP} \Delta_{t\omega}^{\uparrow} - \lambda_{t\omega}^{DA} \Delta_{t\omega}^{\downarrow}) \right] \quad (1)$$

- **Constraints**

$$0 \leq p_t^{DA} \leq P^{nom} \quad \forall t \in T \quad (2)$$

$$\Delta_{t\omega}^{\uparrow} - \Delta_{t\omega}^{\downarrow} = p_{t\omega}^{real} - p_t^{DA} \quad \forall t \in T, \forall \omega \in \Omega \quad (3)$$

$$0 \leq \Delta_{t\omega}^{\uparrow} \leq P^{nom} \quad \forall t \in T, \forall \omega \in \Omega \quad (4)$$

$$0 \leq \Delta_{t\omega}^{\downarrow} \leq P^{nom} \quad \forall t \in T, \forall \omega \in \Omega \quad (5)$$

(2)

Assumptions and considerations

Under the two-price scheme, the settlement price depends on the interaction between the system state $SI_{t\omega} \in \{0, 1\}$, which evolves across hours and scenarios, and the direction of the wind farm's own imbalance. Specifically, when the wind farm's

deviation is beneficial to the system, that is, excess production during a system deficit ($SI_{t\omega} = 1$) or deficit production during a system surplus ($SI_{t\omega} = 0$), it settles at the day-ahead price $\lambda_{t\omega}^{DA}$. Conversely, when the deviation is harmful to the system, excess production during a surplus or deficit production during a deficit, it settles at the balancing price $\lambda_{t\omega}^{BP}$, which penalizes the wind farm. This asymmetry incentivizes the wind farm to submit a more accurate day-ahead offer, since imbalances that worsen the system state are financially penalized, in contrast to the one-price scheme of Task 1.1 where the same balancing price applies regardless of whether the deviation is beneficial or harmful.

Results

The optimal hourly DA offers under the two-price scheme differ from the one-price case. In contrast to the extreme bidding behavior previously observed, in terms of either max capacity or nothing, the producer now tends to bid close to maximum capacity almost all hours of the day. This is due to the difference in payment in the new price scheme. This price scheme penalizes imbalances with the imbalance price, while the beneficial balances are awarded, but only at the DA price. It results in the producer not having any motivation to try and guess the outcome. Since the largest revenue (and thereby profits) are made from producing as much as possible, this is the offer strategy and then the imbalance penalty will be solved regardless in the balancing market. The resulting optimal offers are therefore mainly at 500 MW, with only a few hours showing a slightly reduced bids. The results are presented in Table 2

Hour	1	2	3	4	5	6	7	8	9	10	11	12
Offer [MW]	500	500	500	500	500	418.04	500	408.61	500	500	500	500
Hour	13	14	15	16	17	18	19	20	21	22	23	24
Offer [MW]	500	500	500	500	500	500	500	500	500	500	500	500

Table 2: Optimal offers for each hour under the two-price scheme

Using this bidding strategy, wind generation is still assumed to have a marginal cost of zero, implying that all revenues correspond directly to profits. The distribution of profits across all simulated scenarios under the two-price scheme is illustrated in Figure 2. Compared to the one-price scheme, the profit distribution exhibits a more concentrated shape with a reduced upper range of outcomes.

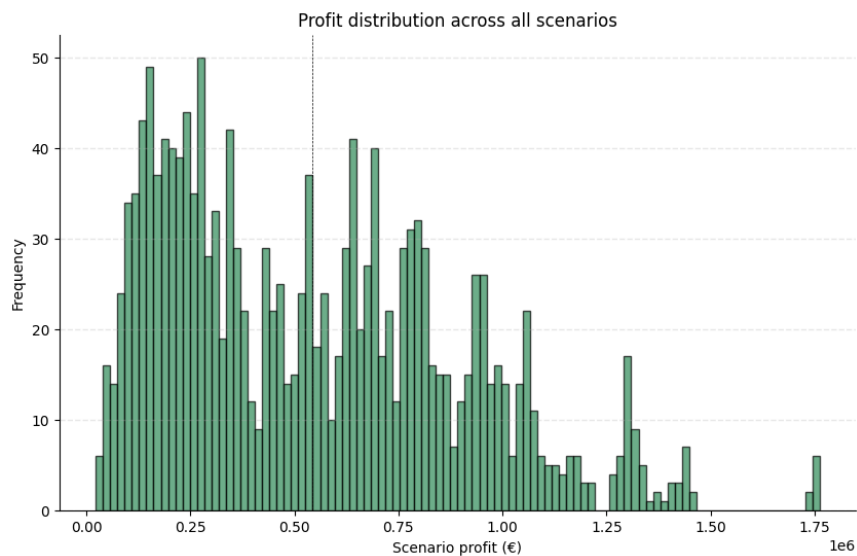


Figure 2: Distribution of profits across all scenarios w. two price scheme

The profit distribution shows a broad spread across scenarios, with values ranging from close to 0 up to approximately 1.75€ million. The distribution appears to have a high concentration of outcomes in the a lower profit range, particularly between roughly 0.1€ million and 0.6€ million, and a gradually decreasing frequency for higher profit levels.

The profit distributions of the two schemes appear similar despite the very different optimal offer strategies. This is explained by the nature of the two-price settlement: by bidding close to full capacity in almost every hour, the wind farm collects most of its revenue directly from the DA market, with balancing deviations playing a secondary role. As a result, profits in both cases are primarily driven by wind production variability across scenarios, producing comparable distribution shapes. The key structural difference, however, is that the two-price scheme eliminates the possibility of exploiting the balancing market opportunistically, which is reflected in the absence of the extreme bidding behavior seen in Task 1.1.

Task 1.3

To assess whether the in-sample offering decisions generalize to unseen scenarios, an 8-fold cross-validation is performed over the full set of $|\Omega| = 1,600$ scenarios, following the procedure outlined in Lecture 8. The 1,600 scenarios are divided into 8 equally-sized folds of 200 scenarios each. In each run $k \in \{1, \dots, 8\}$, one fold (200 scenarios) is used as the in-sample set to solve the optimization problem and obtain the offering decisions $\{p_t^{DA,k}\}_{t \in T}$, while the remaining 7 folds (1,400 scenarios) form the out-of-sample set. The procedure is repeated for both the one-price and two-price schemes. The three steps of the analysis follow Lecture 8 directly:

- **Step 1 (in-sample):** Solve the stochastic offering problem on the 200 in-sample scenarios to obtain the optimal day-ahead offer $\{p_t^{DA,k}\}$ and the corresponding in-sample expected profit.
- **Step 2 (out-of-sample imbalances):** For each of the 1,400 out-of-sample scenarios, fix the offer $\{p_t^{DA,k}\}$ obtained in Step 1 and compute the resulting imbalances and per-scenario profits without re-optimizing.
- **Step 3 (out-of-sample profit):** The out-of-sample expected profit is computed as the average profit across the 1,400 out-of-sample scenarios from Step 2, with the day-ahead offers fixed from Step 1. This quantity is compared against the in-sample expected profit from Step 1 to assess whether the offering decisions generalize to unseen scenarios. A small gap between the two indicates low optimism bias and good out-of-sample performance.

By rotating the in-sample fold across all 8 runs and averaging, any bias introduced by a particular selection of in-sample scenarios is eliminated.

Results

Table 3 reports the in-sample (IS) and out-of-sample (OOS) expected profits for each fold and both settlement schemes, together with the cross-validated averages.

The cross-validated averages are:

- **One-price scheme:** average IS = 532,751 €, average OOS = 532,751 € with no difference between IS and OOS average profits.
- **Two-price scheme:** average IS = 539,348 €, average OOS = 540,692 € so OOS marginally exceeds IS, indicating no optimism bias.

Figure 3 shows the IS and OOS expected profits per fold for both schemes, with dashed lines indicating the cross-validated averages. Figure 4 shows the day-ahead offers $p_t^{DA,k}$ across all 8 folds and 24 hours.

The cross-validation results show that IS and OOS profits match closely under both schemes. For the one-price scheme the gap is exactly zero, while for the two-price scheme the OOS profit marginally exceeds the IS profit by € 1,344 confirming no optimism bias and that the offering decisions generalize well with no sign of overfitting. The fold-level swings visible

Table 3: 8-fold cross-validation results: IS and OOS expected profits under the one-price and two-price schemes.

Fold	One-price (€)		Two-price (€)	
	IS	OOS	IS	OOS
1	195,562	580,921	239,677	585,185
2	835,877	489,447	809,836	500,707
3	717,836	506,310	700,595	516,552
4	805,932	493,725	781,590	505,020
5	396,163	552,264	415,450	558,120
6	227,971	576,291	271,793	580,602
7	176,626	583,626	222,518	587,579
8	906,040	479,424	873,325	491,768
Mean	532,751	532,751	539,348	540,692



Figure 3: Task 1.3 - IS vs. OOS expected profit per fold under the one-price scheme (left) and the two-price scheme (right). Dashed lines indicate the cross-validated averages.

in Figure 3 are most likely due to uneven price scenario distributions within each small in-sample set of 200.

Regarding the offer profiles in Figure 4, the one-price scheme maintains its all-or-nothing behavior identically across all folds, while the two-price scheme stays close to 500 MW for nearly all hours, with only a brief dip around hour 4. The two-price scheme also edges ahead in average profitability (€539,348 vs €532,751), a structural advantage that holds out-of-sample. Given that the optimism gap is already essentially zero with 200 scenarios, there is no strong case for modifying the in-sample size. It is worth noting, however, that the robustness of the one-price solution is largely a consequence of its corner structure, so the all-or-nothing bidding policy is insensitive to the specific scenario sample by design, which means the cross-validation result reflects the simplicity of the decision rather than the richness of the data.

Task 1.4

Following Lecture 9, the risk-neutral stochastic offering models developed in Tasks 1.1 and 1.2 are extended to account for risk aversion using the Conditional Value-at-Risk (CVaR) at confidence level $\alpha = 0.90$. The aim is to protect the

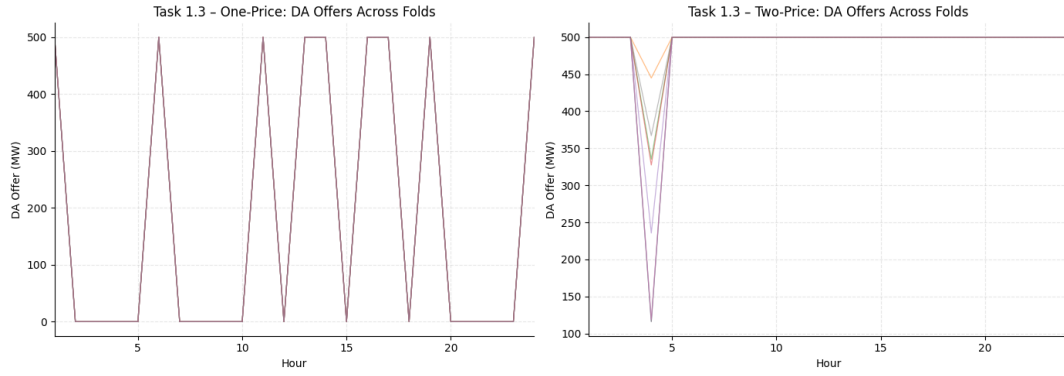


Figure 4: Task 1.3 - Day-ahead offers across all 8 folds under the one-price scheme (left) and the two-price scheme (right).

wind producer against adverse low-profit outcomes while retaining incentives from expected profit maximization. As in Tasks 1.1 and 1.2, the day-ahead offer p_t^{DA} is a *first-stage* (here-and-now) decision: it must be submitted before any scenario realization is observed and is therefore *not* indexed by ω . The imbalance variables $\Delta_{t\omega}^\uparrow$ and $\Delta_{t\omega}^\downarrow$ remain *second-stage* (wait-and-see) decisions indexed by ω , since they are determined only after the realization of wind power production and prices is known. The total profit realized in scenario ω is the sum of hourly profits across all time periods:

$$\Pi_\omega = \sum_{t \in T} \text{Profit}_{t\omega} \quad \forall \omega \in \Omega \quad (1)$$

where $\text{Profit}_{t\omega}$ is defined by the applicable settlement scheme (one-price or two-price), as specified in the Profit Functions paragraph below.

Following Rockafellar & Uryasev, two auxiliary variables are introduced to linearize the CVaR:

- $\zeta \in \mathbb{R}$: the VaR threshold, determined by the optimization (a variable, not a fixed parameter);
- $\eta_\omega \geq 0$, for all $\omega \in \Omega$: the profit shortfall below ζ in scenario ω , i.e. $\eta_\omega = \max\{0, \zeta - \Pi_\omega\}$, enforced through linear constraints (10) and (11).

The CVaR is then expressed as:

$$\text{CVaR}_\alpha(\Pi) = \zeta - \frac{1}{1-\alpha} \sum_{\omega \in \Omega} \pi_\omega \eta_\omega \quad (2)$$

Mean-CVaR objective

Rather than maximizing expected profit alone, the risk-averse objective maximizes a weighted combination of expected profit and CVaR:

$$\max \underbrace{\sum_{\omega \in \Omega} \pi_\omega \Pi_\omega}_{E[\Pi]} + \beta \underbrace{\left(\zeta - \frac{1}{1-\alpha} \sum_{\omega \in \Omega} \pi_\omega \eta_\omega \right)}_{\text{CVaR}_\alpha(\Pi)} \quad (3)$$

where $\beta \geq 0$ is the risk-aversion weight, set by the decision maker. Setting $\beta = 0$ recovers the risk-neutral formulations of Tasks 1.1 and 1.2 exactly.

Complete risk-averse optimization model.

Substituting (2) into (3) and appending all operational and CVaR constraints yields the full linear program:

• Objective function

$$\max_{p_t^{DA}, \Delta_{t\omega}^\uparrow, \Delta_{t\omega}^\downarrow, \zeta, \eta_\omega} \sum_{\omega \in \Omega} \pi_\omega \Pi_\omega + \beta \left(\zeta - \frac{1}{1-\alpha} \sum_{\omega \in \Omega} \pi_\omega \eta_\omega \right) \quad (4)$$

• **Constraints**

$$\Pi_\omega = \sum_{t \in T} \text{Profit}_{t\omega} \quad \forall \omega \in \Omega \quad (5)$$

$$0 \leq p_t^{DA} \leq P^{\text{nom}} \quad \forall t \in T \quad (6)$$

$$\Delta_{t\omega}^\uparrow - \Delta_{t\omega}^\downarrow = p_{t\omega}^{\text{real}} - p_t^{DA} \quad \forall t \in T, \forall \omega \in \Omega \quad (7)$$

$$0 \leq \Delta_{t\omega}^\uparrow \leq P^{\text{nom}} \quad \forall t \in T, \forall \omega \in \Omega \quad (8)$$

$$0 \leq \Delta_{t\omega}^\downarrow \leq P^{\text{nom}} \quad \forall t \in T, \forall \omega \in \Omega \quad (9)$$

$$\eta_\omega \geq \zeta - \Pi_\omega \quad \forall \omega \in \Omega \quad (10)$$

$$\eta_\omega \geq 0, \quad \forall \omega \in \Omega \quad (11)$$

Constraints (6)–(9) are identical to those of Tasks 1.1 and 1.2, with the imbalance variables bounded above by P^{nom} to prevent artificial inflation of simultaneous upward and downward deviations. Constraints (10) and (11) together enforce $\eta_\omega = \max\{0, \zeta - \Pi_\omega\}$, linearizing the $\max[\cdot]$ operator in the CVaR definition. Note that ζ is a free variable determined endogenously: the optimizer sets it to the $(1-\alpha)$ -quantile of the profit distribution, which is well-defined because the profit is bounded above by P^{nom} through constraints (8) and (9).

Profit functions

The per-hour profit $\text{Profit}_{t\omega}$ entering constraint (5) is defined according to the settlement scheme in use.

One-price scheme (Task 1.1):

$$\text{Profit}_{t\omega} = \lambda_{t\omega}^{DA} p_t^{DA} + \lambda_{t\omega}^{BP} (\Delta_{t\omega}^\uparrow - \Delta_{t\omega}^\downarrow). \quad (12)$$

Two-price scheme (Task 1.2):

$$\text{Profit}_{t\omega} = \lambda_{t\omega}^{DA} p_t^{DA} + SI_{t\omega} (\lambda_{t\omega}^{DA} \Delta_{t\omega}^\uparrow - \lambda_{t\omega}^{BP} \Delta_{t\omega}^\downarrow) + (1 - SI_{t\omega}) (\lambda_{t\omega}^{BP} \Delta_{t\omega}^\uparrow - \lambda_{t\omega}^{DA} \Delta_{t\omega}^\downarrow). \quad (13)$$

Interpretation of β

The risk-aversion weight β controls the balance between expected profit and tail protection:

- $\beta = 0$: *risk-neutral* case; the model reduces to the expected-profit maximization of Tasks 1.1 and 1.2.
- $\beta > 0$: *risk-averse* case; the optimizer sacrifices some expected profit to raise the average profit in the worst 10% of scenarios.
- $\beta \rightarrow \infty$: the solution converges to *pure CVaR maximization*, concentrating entirely on the worst-case tail.

Results

The risk-averse offering strategy was evaluated under both the one-price and two-price balancing schemes for increasing values of the risk-aversion parameter $\beta \in \{0, 0.1, 0.3, 0.5, 0.75, 1, 1.5, 2, 3, 5\}$, with CVaR confidence level $\alpha = 0.90$. The complete numerical results are summarized in Table 4, while the efficient frontier is shown in Figure 5.

Table 4: Task 1.4 — Expected profit and CVaR_{0.90} under one-price and two-price schemes for varying β .

β	One-price		Two-price	
	$E[\Pi]$ (EUR/day)	CVaR _{0.90} (EUR/day)	$E[\Pi]$ (EUR/day)	CVaR _{0.90} (EUR/day)
0.00	532,751	49,718	459,333	−1,078
0.10	532,751	49,718	459,156	3,306
0.30	532,751	49,718	458,218	8,468
0.50	532,751	49,718	456,398	13,315
0.75	532,751	49,718	455,460	14,954
1.00	532,751	49,718	453,798	16,841
1.50	532,751	49,718	449,906	19,892
2.00	532,751	49,718	448,812	20,488
3.00	532,751	49,718	448,695	20,545
5.00	532,751	49,718	448,695	20,545

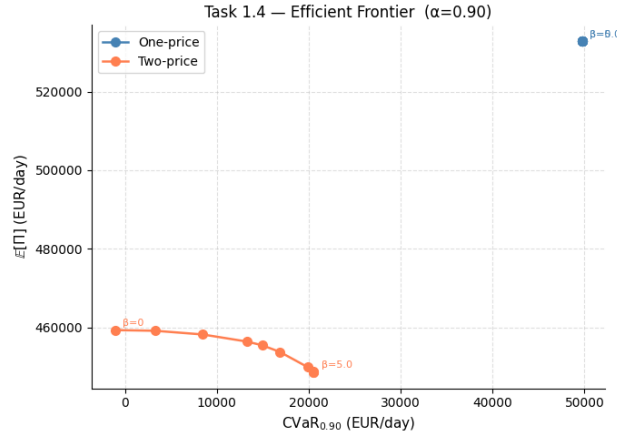


Figure 5: Task 1.4 — Efficient frontier ($\alpha = 0.90$) for one-price and two-price schemes across all values of β .

Under the one-price scheme, the risk-neutral solution ($\beta = 0$) already shows corner bidding: the model bids either $P^{\text{nom}} = 500$ MW or 0 MW in every hour. This happens because, under the one-price scheme, the per-hour profit is linear in p_t^{DA} with coefficient $E[\lambda_t^{DA} - \lambda_t^{BP}]$. Since this is a linear program, the optimal solution always lies at a bound: the model bids P^{nom} at hours where $E[\lambda_t^{DA} - \lambda_t^{BP}] > 0$ (9 hours), and bids 0 at hours where this quantity is negative (15 hours), independently for each hour. As a consequence, increasing β does not change the offering strategy at all: both $E[\Pi]$ and CVaR_{0.90} remain constant at 532,751 and 49,718 EUR/day respectively for all β , as seen in Figure 6. The efficient frontier in Figure 5 degenerates to a single point, confirming that, within this system, risk aversion has no effect. It should be noted, however, that this result is partly an artifact of the modeling assumptions. The model implicitly assumes that the producer always participates in the balancing market, since actual wind production almost never exactly equals the DA offer.

Under the two-price scheme, the model has genuine flexibility to trade expected profit for improved worst-case outcomes. At $\beta = 0$, the risk-neutral solution yields $E[\Pi] = 459,333$ EUR/day with a negative CVaR of $-1,078$ EUR/day, meaning the worst 10% of scenarios result in daily losses on average. This occurs because the two-price asymmetry penalizes deviations that go against the system state: over-production during system surplus is settled at $\lambda^{BP} = 0.85 \lambda^{DA}$ rather than the full DA price, and under-production during system deficit incurs a cost at $\lambda^{BP} = 1.25 \lambda^{DA}$. As β increases, CVaR improves rapidly at first and then saturates around $\beta \approx 3$, beyond which no further improvement is achieved. By

$\beta = 5$, the expected profit decreases by only 2.32% to 448,695 EUR/day, while CVaR improves by +21,623 EUR/day to 20,545 EUR/day. The convex shape of the two-price efficient frontier in Figure 5 clearly illustrates this classic risk-return trade-off.

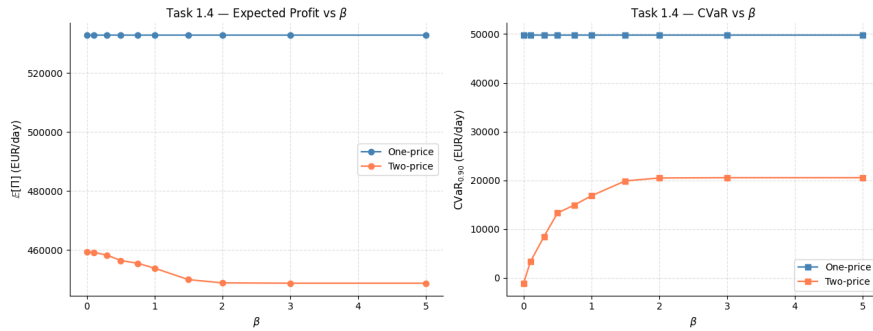


Figure 6: Task 1.4 - Expected profit $E[\Pi]$ (left) and $CVaR_{0.90}$ (right) as a function of β for one-price and two-price schemes.

Offering strategy

Figure 7 shows how day-ahead offers evolve with β under both schemes. Under the one-price scheme, the all-or-nothing bidding pattern is completely invariant to β : all three curves ($\beta = 0, 1, 5$) are identical, alternating between 0 and 500 MW hour by hour. Under the two-price scheme, offers are more moderate and decrease systematically as β grows, reflecting the model's effort to reduce exposure to harmful imbalances in the worst scenarios. Hours with high expected wind production retain non-zero offers even at large β , while more uncertain hours are progressively driven toward zero.

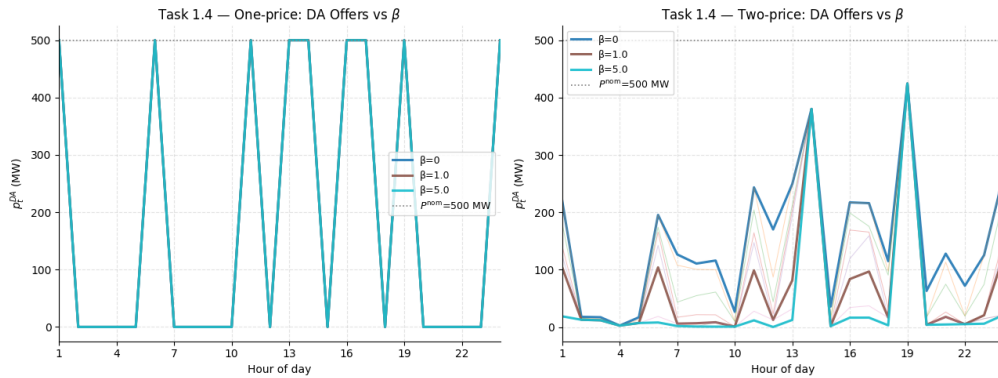


Figure 7: Task 1.4 - Optimal day-ahead offers p_t^{DA} as a function of hour for selected values of β , under one-price (left) and two-price (right) schemes.

Profit distributions

The profit histograms in Figure 8 confirm the above findings. Under the one-price scheme, the distribution is entirely unchanged across all β values, consistent with the flat efficient frontier. Under the two-price scheme, increasing β progressively compresses the left tail of the distribution: the VaR shifts rightward from 18,977 EUR at $\beta = 0$ to 43,913 EUR at $\beta = 5$, confirming that risk aversion effectively protects against the worst outcomes. The right tail is also slightly truncated due to the more conservative DA offers, but the overall cost in expected profit remains small.

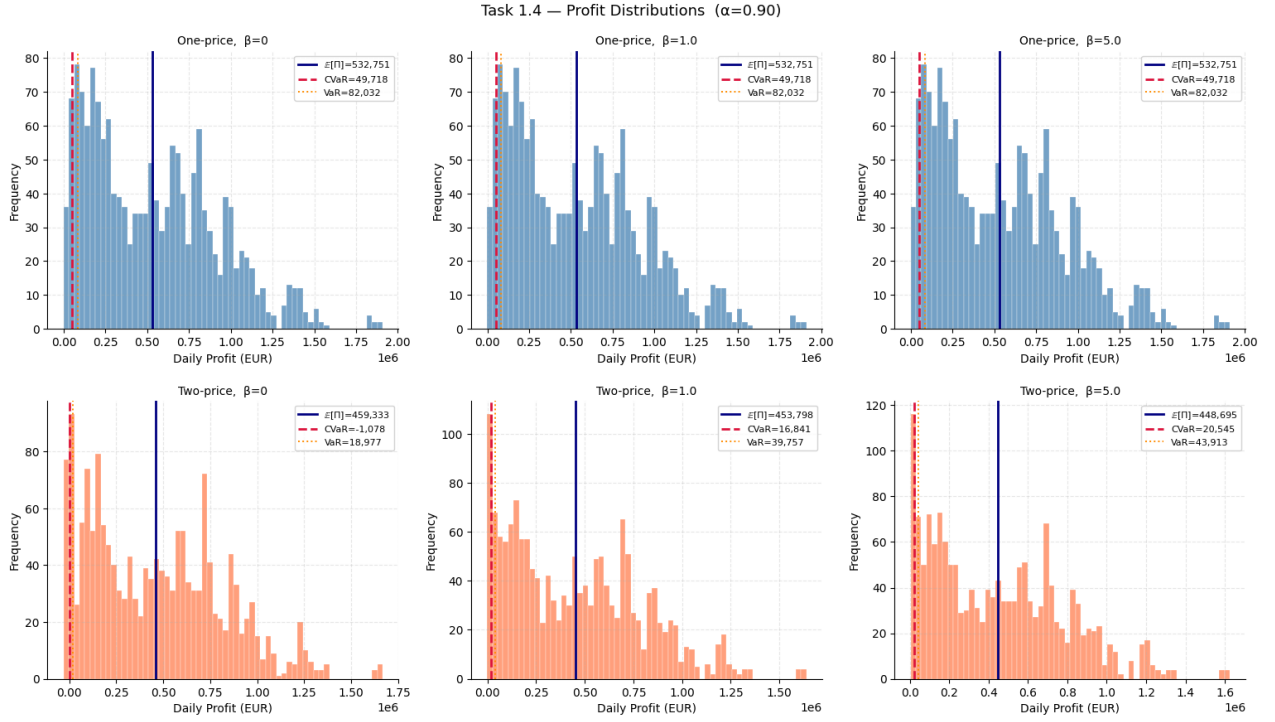


Figure 8: Task 1.4 — Profit distributions across 1,600 scenarios for $\beta \in \{0, 1, 5\}$ under one-price (top row) and two-price (bottom row) schemes. Vertical lines indicate $E[\Pi]$, $CVaR_{0.90}$, and $VaR_{0.90}$.

Sensitivity to in-sample size

Since the one-price optimal policy is determined solely by the sign of $E[\lambda_t^{DA} - \lambda_t^{BP}]$ per hour, the offering decisions are structurally insensitive to the specific scenario sample drawn. As a result, varying the in-sample size does not meaningfully affect solution quality under the one-price scheme. For the two-price scheme, where offers are more moderate and scenario-dependent, a larger in-sample set could in principle reduce the variance of the optimal offers across folds, though the cross-validation results of Task 1.3 already suggest that 200 scenarios are sufficient to produce stable decisions.

Step 2 - Participation in Ancillary Service Markets

The first step in Task 2 is to generate 300 stochastic load profiles in order to assess the pricing strategies in a grid with a flexible load. The load profiles represent a single bidding hour with minute-by-minute resolution. Consequently, each profile contains 60 observations which must satisfy the following consumption and ramping constraints:

$$220 \leq L_m \leq 600, \quad |L_m - L_{m-1}| \leq 35 \quad \forall m \in \mathcal{M},$$

where $\mathcal{M} = \{1, \dots, 60\}$ denotes the set of minutes within the bidding hour. The purpose of imposing these constraints on the generated load profiles is to ensure realistic operating conditions and thereby mimic real-life consumption behavior.

Through a Python simulation, the stochastic load profiles were generated as bounded random profiles. First, the load was sampled uniformly within the interval $[220, 600]$ kW. This process was repeated for each minute of the one-hour period, where the next value was sampled uniformly from the feasible interval

$$[\max\{220, L_{m-1} - 35\}, \min\{600, L_{m-1} + 35\}].$$

By constructing the code in this manner, it is ensured that all constraints are satisfied at each timestep.

Table 5: Load profile generation parameters

Parameter	Value	Unit	Description
Number of profiles	300	-	Total number of generated scenarios
Time resolution	1	min	60 values per profile
Profile duration	1	hour	One bidding hour
Load range	220–600	kW	Feasible consumption interval
Max. minute change	35	kW	Ramp constraint
In-sample profiles	100	-	Used for optimization
Out-of-sample profiles	200	-	Used for validation
Random seed	42	-	Reproducibility

Out of the 300 generated profiles, 100 were selected as in-sample profiles used to determine the reserve bid, while the remaining 200 profiles were used as out-of-sample profiles for validation of the obtained solution.

Task 2.1: In-sample Decision Making Under the P90 Requirement

Based upon the fomulation received in lectur 10, the FCR-D UP bidding problem can be modeled as a joint chance-constrained program. Here, $c^\uparrow$ denotes the reserve capacity bid of FCR-D to the given hour. Additionally, for a flexible load, we consider the available upward flexibility $F_m^\uparrow$:

$$F_m^\uparrow = L_m - 220,$$

For this assignment, the minimum feasible consumption level is 220 kW. If the load falls below this value, the flexible load is considered unable to operate. The next step is therefore to define the P90 requirement, which will subsequently be tested against each obtained solution:

$$\max_{c^\uparrow \geq 0} c^\uparrow$$

subject to

$$P\left(c^\uparrow \leq F_m^\uparrow, \forall m \in \mathcal{M}\right) \geq 1 - \varepsilon,$$

where $\varepsilon = 0.1$ means that our reserve bid must be with at least 90% probability of being feasible across all minutes.

Using the 100 in-sample profiles, the optimal FCR-D UP reserve bid is determined using both ALSO-X and CVaR for the chance-constrained optimization problem. By testing both methods, it becomes possible to compare how the two approximations perform when solving the chance-constrained problem. Both approaches have their respective advantages: ALSO-X seeks to limit the number of violations, whereas CVaR controls the magnitude of the worst-case shortfalls.

ALSO-X formulation

$$\max_{c^\uparrow \geq 0, y_{m,w}} c^\uparrow$$

subject to

$$c^\uparrow - F_{m,w}^\uparrow \leq M y_{m,w}, \quad \forall m \in \mathcal{M}, \forall w \in W \quad (14)$$

$$\sum_{m \in \mathcal{M}} \sum_{w \in W} y_{m,w} \leq q$$

$$y_{m,w} \in \{0, 1\}, \quad \forall m \in \mathcal{M}, \forall w \in W$$

CVaR formulation

$$\max_{c^\uparrow \geq 0, \beta \leq 0, \zeta_{m,w}} c^\uparrow$$

subject to

$$\zeta_{m,w} \geq c^\uparrow - F_{m,w}^\uparrow - \beta, \quad \forall m \in \mathcal{M}, \forall w \in W \quad (15)$$

$$\beta + \frac{1}{q} \sum_{m \in \mathcal{M}} \sum_{w \in W} \zeta_{m,w} \leq 0$$

$$\zeta_{m,w} \geq 0, \quad \forall m \in \mathcal{M}, \forall w \in W$$

Each approximation method was used to compute and approximate the reserve bid. The in-sample results for both computations can be found in Table 6.

Table 6: In-sample results for Task 2.1

Method	Optimal bid [kW]	Violation rate [%]
ALSO-X	45.03	10.00
CVaR	24.26	4.83

From the presented results, ALSO-X generates the highest reserve bid and exhibits a significantly higher violation rate of 10.00%. In comparison, the CVaR solution appears more conservative, with approximately half the bid size of ALSO-X and a violation rate of only 4.83%. In the model, the maximum allowable violation rate is set to $\varepsilon = 10\%$, which corresponds exactly to the result obtained using ALSO-X. This may explain why the method is able to identify such a high feasible reserve bid. In contrast, the CVaR formulation does not explicitly count the number of violations. Instead, it penalizes the magnitude of violations through the expected shortfall in the worst ε -fraction of scenarios. Consequently, this leads to a considerably more conservative reserve bid for the CVaR approach.

From the results, it can be concluded that there are relatively large differences between the two approximations. ALSO-X yields the largest reserve bid under the specified constraints and requirements, whereas CVaR provides a more conservative reserve bid with a substantially larger safety margin.

Task 2.2: Verification of the P90 Requirement Using Out-of-Sample Analysis

In the previous task, the 100 in-sample profiles were investigated using the P90 requirement. In this task, the obtained reserve bids are evaluated against the 200 out-of-sample profiles. Therefore, it is not necessary to further define any mathematical formulations or perform additional optimizations, as the same approximation methods from Equations 14 and 15 are used.

However, the results must now be compared against the minute-level reserve availability, where a reserve shortfall occurs whenever

$$F_m^\uparrow < c^\uparrow.$$

As in the previous task, the P90 requirement is considered satisfied as long as the reliability remains at or above 90%.

Each approximation method was used to compute and approximate the reserve bid. The out-of-sample results for both computations are summarized in Table 7.

Table 7: Out-of-sample verification results for Task 2.2

Method	Bid (kW)	Reliability (%)	Shortfall rate (%)	Meets P90
ALSO-X	45.03	91.63	8.37	Yes
CVaR	24.26	96.25	3.75	Yes

If we first consider the ALSO-X solution, the out-of-sample analysis results in 1004 shortfalls out of 12000 minute-level observations, where the average shortfall magnitude is 19.81 kW and the maximum shortfall is 45.00 kW. In comparison, the CVaR solution results in 450 shortfalls out of 12000 observations, where the average shortfall magnitude is 11.13 kW and the maximum shortfall is 24.23 kW.

Furthermore, the P90 requirement is clearly satisfied by both approximation methods, as both achieve a reliability above 90%. Similar to Task , ALSO-X provides the larger reserve bid, but due to the higher shortfall rate, it also becomes more risky than the CVaR approach. In contrast, CVaR remains more conservative while simultaneously achieving a higher reliability because of the lower number of shortfalls. The results indicate that both approximation methods

perform well when evaluated on the out-of-sample profiles. Since the violation fraction was originally selected to satisfy the P90 requirement, the obtained reserve bids remain feasible without requiring re-optimization. ALSO-X, by fully utilizing its violation budget in-sample, operates closer to the constraint boundary and consequently experiences a higher out-of-sample shortfall rate. CVaR, being structurally more conservative, maintains a larger safety margin and therefore achieves stronger out-of-sample reliability. From a grid reliability perspective, CVaR can therefore be considered the more robust solution, whereas ALSO-X remains economically attractive due to its higher reserve bid.

Task 2.3: Energinet Perspective

This task investigates the impact of modifying the P90 requirement from Energinet's perspective. The reliability threshold is varied between 80% and 100% using the ALSO-X approximation method, while the CVaR approximation is omitted from this analysis. The optimal reserve bids are determined using the in-sample profiles and subsequently evaluated on the out-of-sample profiles.

From the previous analyses, a clear relationship between the reliability requirement and the optimal reserve bid is observed. Increasing the reliability threshold reduces the optimal reserve bid, whereas relaxing the requirement allows for larger reserve bids. For example, the optimal reserve bid decreases from 82.48 kW at an 80% reliability threshold to 44.93 kW at the 90% threshold, eventually approaching zero at the 100% threshold. Simultaneously, the out-of-sample shortfall decreases as the reliability requirement becomes stricter, indicating improved delivery performance and system reliability at the expense of reduced reserve provision.

This relationship is illustrated in Figure 9 and Table 8. The left figure shows how the optimal reserve bid varies with the reliability threshold, while the right figure illustrates the expected out-of-sample shortfall and thereby highlights the trade-off between reserve provision and delivery reliability.

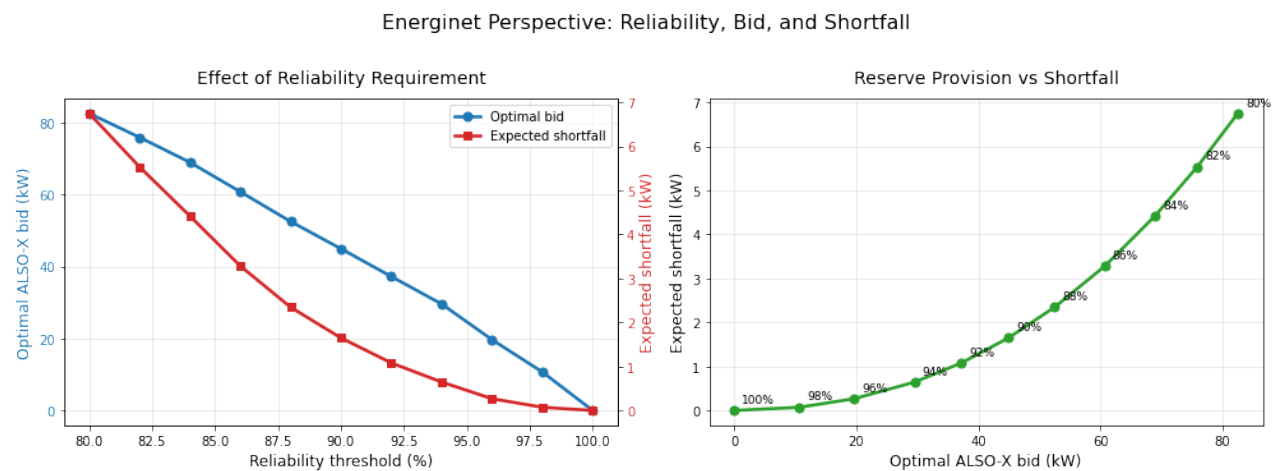


Figure 9: Effect of the reliability threshold on the optimal ALSO-X reserve bid and the expected out-of-sample shortfall. A stricter P-threshold reduces both the reserve bid and the expected shortfall, illustrating the trade-off between reserve provision and delivery reliability.

P-threshold (%)	Bid (kW)	Reliability (%)	Shortfall (%)	Exp. shortfall (kW)
80	82.48	80.57	19.43	6.75
82	75.84	82.80	17.20	5.53
84	68.91	84.88	15.12	4.42
86	60.74	87.38	12.63	3.28
88	52.48	89.77	10.23	2.35
90	44.93	91.66	8.34	1.65
92	37.22	93.50	6.50	1.08
94	29.59	95.18	4.83	0.65
96	19.69	97.13	2.88	0.27
98	10.72	98.59	1.41	0.07
100	0.04	99.98	0.02	0.00

Table 8: Effect of reliability threshold on ALSO-X bid and out-of-sample performance

The results clearly illustrate the trade-off between reserve provision and delivery reliability. Increasing the reliability requirement improves system security and reduces reserve shortfalls, but simultaneously lowers the reserve capacity that can be offered. Conversely, relaxing the reliability requirement enables larger reserve bids at the expense of higher shortfall rates. From Energinet's perspective, maintaining a stable and secure power system is the primary objective, while still ensuring sufficient reserve availability. However, if the reliability requirement becomes too strict, the reserve volume decreases significantly. In the limiting case of a 100% reliability requirement, almost no reserve can be offered, since the reserve bid must remain feasible in every realization. Conversely, at the 80% threshold, substantially larger reserve bids can be offered, although with noticeably higher shortfall rates. Overall, the analysis demonstrates that tightening the P-threshold improves delivery reliability and reduces reserve shortfalls, but at the expense of reduced reserve provision. Energinet must therefore balance delivery reliability against reserve capacity through the reliability requirement.

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Appendix